

Observation on Existing Work and Proposed Improvement

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1.1 Observation from Existing Work

The existing work follows a data-driven approach in which flow visualization images are first processed using a convolutional autoencoder. The autoencoder compresses each input flow image into a low-dimensional latent representation. Sparse Identification of Nonlinear Dynamics (SINDy) is then applied to these latent variables to identify the governing dynamics.

The general pipeline of the existing method can be written as:

$$\text{Flow image} \rightarrow \text{Autoencoder} \rightarrow \text{Latent variables} \rightarrow \text{SINDy} \quad (1)$$

If the input image at time t is denoted by $I(x, y, t)$, the encoder maps it to a latent vector:

$$I(x, y, t) \rightarrow \mathbf{z}(t) \quad (2)$$

where

$$\mathbf{z}(t) = [z_1(t), z_2(t), z_3(t), \dots, z_n(t)] \quad (3)$$

SINDy is then used to identify equations of the form:

$$\dot{\mathbf{z}} = \Theta(\mathbf{z})\Xi \quad (4)$$

where $\Theta(\mathbf{z})$ is the candidate library and Ξ contains the sparse coefficients.

Although this approach is useful for learning low-dimensional dynamics, the identified equations are latent-space equations rather than physical flow governing equations. The variables $z_1, z_2, z_3, \dots$ do not directly represent physical quantities such as velocity, pressure, vorticity, or strain rate.

1.2 Limitations of the Existing Work

The main limitation of the existing work is that it attempts to discover governing equations from compressed image features instead of physical flow variables. As a result, the identified equation is of the form:

$$\dot{z}_i = f(z_1, z_2, z_3, \dots, z_n) \quad (5)$$

rather than a physical partial differential equation such as:

$$u_t + uu_x + vv_y = -\frac{1}{\rho}p_x + \nu(u_{xx} + u_{yy}) \quad (6)$$

or

$$v_t + uv_x + vv_y = -\frac{1}{\rho}p_y + \nu(v_{xx} + v_{yy}) \quad (7)$$

The following limitations were observed:

1. The model uses grayscale image intensity $I(x, y, t)$ instead of physical velocity fields $u(x, y, t)$ and $v(x, y, t)$.
2. The autoencoder produces latent variables that are not directly interpretable as physical flow quantities.
3. Spatial derivatives such as u_x , u_y , u_{xx} , u_{yy} , v_x , and v_y are not explicitly computed.
4. The SINDy candidate library is built using latent variables rather than physical PDE terms.
5. The identified equations describe latent dynamics, not real governing equations of fluid flow.
6. The method may identify one simple equation correctly, but it fails to generalize to multiple physical equations because the required physical information is missing.
7. The approach is sensitive to noise, frame quality, time-step selection, and sparsity threshold values.
8. The method does not include high-performance computing acceleration for large-scale video processing, derivative computation, or model training.

1.3 Reason for Partial Success

The existing method may correctly identify one equation because the corresponding latent variable may have simple dynamics. However, this does not guarantee that the discovered equation represents the actual physical law. Since the latent variables are

learned internally by the neural network, their relationship with real physical variables is unclear.

Therefore, even if one latent equation appears correct, the method cannot reliably recover the full set of physical governing equations for fluid flow.

1.4 Proposed Target for Improvement

The objective of the proposed work is to move from latent-space equation discovery to physical-space governing equation discovery. Instead of applying SINDy directly to autoencoder latent variables, the proposed method aims to extract physical flow fields from flow visualization images or videos.

The improved pipeline is proposed as:

$$\text{Flow video} \rightarrow \text{Frame extraction} \rightarrow \text{Optical flow/PIV} \rightarrow u(x, y, t), v(x, y, t) \quad (8)$$

After obtaining the velocity fields, physical derivatives will be computed:

$$u_t, \quad u_x, \quad u_y, \quad u_{xx}, \quad u_{yy} \quad (9)$$

$$v_t, \quad v_x, \quad v_y, \quad v_{xx}, \quad v_{yy} \quad (10)$$

These terms will be used to construct a physical PDE candidate library:

$$\Theta = [1, u, v, u_x, u_y, v_x, v_y, uu_x, vu_y, uv_x, vv_y, u_{xx}, u_{yy}, v_{xx}, v_{yy}] \quad (11)$$

SINDy/PDE-FIND will then be applied to identify equations of the form:

$$u_t = \Theta \xi_u \quad (12)$$

$$v_t = \Theta \xi_v \quad (13)$$

where ξ_u and ξ_v are sparse coefficient vectors.

1.5 Role of PINN in the Proposed Work

Physics-Informed Neural Networks (PINNs) will be introduced to improve the quality of the velocity fields and their derivatives. Since optical flow or PIV data may contain noise, direct finite-difference derivatives can be inaccurate. A PINN can be trained to

learn smooth approximations:

$$u_\theta(x, y, t), \quad v_\theta(x, y, t), \quad p_\theta(x, y, t) \quad (14)$$

Using automatic differentiation, the required derivatives can be computed more accurately:

$$\frac{\partial u_\theta}{\partial t}, \quad \frac{\partial u_\theta}{\partial x}, \quad \frac{\partial u_\theta}{\partial y}, \quad \frac{\partial^2 u_\theta}{\partial x^2}, \quad \frac{\partial^2 u_\theta}{\partial y^2} \quad (15)$$

and similarly for v_θ .

Thus, the improved framework becomes:

$$\text{Noisy } u, v \rightarrow \text{PINN smoothing} \rightarrow \text{clean derivatives} \rightarrow \text{SINDy/PDE-FIND} \quad (16)$$

1.6 Role of HPC in the Proposed Work

High-Performance Computing (HPC) will be introduced to improve the scalability and computational efficiency of the proposed framework. The following stages are suitable for HPC acceleration:

1. Parallel frame extraction from flow videos.
2. Parallel optical flow or PIV computation for consecutive frame pairs.
3. Parallel computation of spatial and temporal derivatives.
4. Parallel construction of the PDE candidate library.
5. GPU-accelerated PINN training using PyTorch.
6. Parallel hyperparameter search for SINDy threshold, polynomial degree, PINN architecture, and learning rate.

The HPC-enhanced pipeline can be written as:

$$\text{Video} \rightarrow \text{Parallel preprocessing} \rightarrow \text{Parallel optical flow/PIV} \quad (17)$$

$$\rightarrow \text{GPU-PINN} \rightarrow \text{Parallel PDE library construction} \rightarrow \text{SINDy/PDE-FIND} \quad (18)$$

The performance of the HPC implementation will be evaluated using runtime, speedup, and parallel efficiency:

$$\text{Speedup} = \frac{T_s}{T_p} \quad (19)$$

$$\text{Efficiency} = \frac{\text{Speedup}}{N_p} \quad (20)$$

where T_s is the serial execution time, T_p is the parallel execution time, and N_p is the number of processors or cores.

1.7 Expected Contribution

The expected contribution of the proposed work is to develop an HPC-accelerated physical equation discovery framework for fluid flow data. Unlike the existing latent-space autoencoder-SINDy approach, the proposed method aims to identify interpretable governing equations in terms of physical variables.

The main improvements are summarized as follows:

1. Replacement of latent variables with physical velocity fields $u(x, y, t)$ and $v(x, y, t)$.
2. Extraction of velocity fields from flow visualization images or videos using optical flow/PIV.
3. Computation of physical derivatives required for PDE discovery.
4. Use of PINN for denoising and accurate derivative estimation.
5. Application of SINDy/PDE-FIND to discover physical governing equations.
6. Introduction of HPC acceleration for large-scale video processing and model training.
7. Validation of the discovered equations using reconstruction error, prediction error, and physical residual error.

1.8 Proposed Research Objective

The main objective of this work is:

To develop an HPC-accelerated PINN-SINDy framework for discovering interpretable physical governing equations from flow visualization images or videos by extracting velocity fields, computing physical derivatives, and applying sparse PDE identification.

1.9 Conclusion

The existing work demonstrates the possibility of using autoencoder-based SINDy for learning low-dimensional dynamics from flow images. However, the discovered equations are limited to latent-space dynamics and are not directly interpretable as physical flow governing equations. The proposed work aims to overcome this limitation by extracting physical velocity fields from flow videos, applying PINN-based smoothing, computing physical derivatives, and using SINDy/PDE-FIND to identify real governing equations. HPC will be incorporated to accelerate the computationally expensive stages and make the framework scalable for large flow datasets.